Name: ____________________

Worksheet

Congruent triangles (ASA and AAS)

Task A: Constructing triangles

1) Construct the following triangles:

a) Triangle ABC, where AB = 7.5 cm, $\angle CAB = 54°$ and $\angle ABC = 47°$

b) Triangle DEF, where $\angle DEF = 67°$, $\angle EFD = 25°$ and EF = 8.2 cm

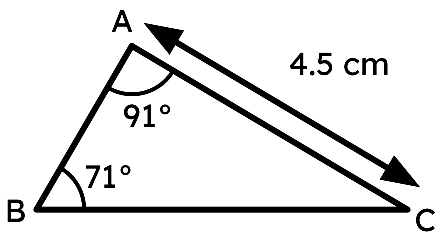
c) Triangle GHI, where GH = 7.1 cm, $\angle GHI = 113°$ and $\angle HIG = 42°$

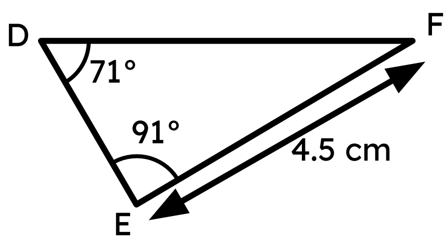
d) Triangle JKL, where ∠JKL = 38°, JL = 5.6 cm and KJ = KL

Task B: Justifying congruence by ASA/AAS

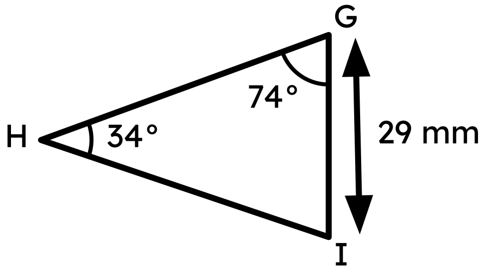
1) Prove each pair of triangles are congruent by ASA/AAS.

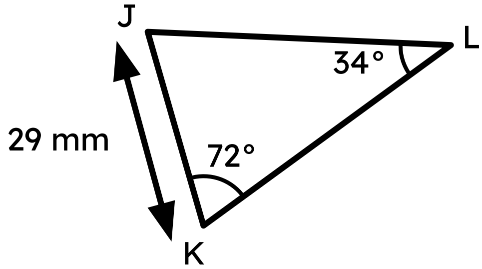
a)



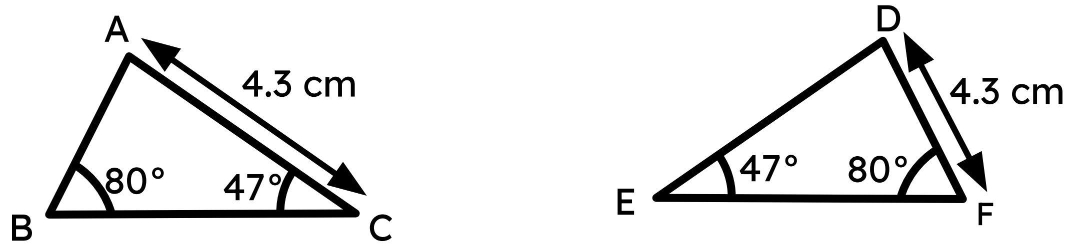


b)





2) Why are these two triangles not congruent by ASA?



3) Given a parallelogram ABCD, prove that triangle ABC is congruent to triangle ACD by ASA/AAS.

4) Point E is the midpoint of AC and BD. Prove that triangle AEB is congruent to triangle ECD by ASA/AAS.